

STUTTGART SCHNARREN, Germany

WMO: 107390

Lat: 48.833N

Lon: 9.200E

Elev: 1033

StdP: 14.16

Time Zone: 1.00 (EUC)

Period: 84-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.3	(c) 15.8	(d) 5.5	(e) 7.6	(f) 14.0	(g) 10.0	(h) 9.5	(i) 17.1	(j) 25.9	(k) 42.3	(l) 21.8	(m) 41.0	(n) 6.4	(o) 110	(p) 0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 16.8	(c) 85.2	(d) 67.3	(e) 82.0	(f) 65.5	(g) 79.0	(h) 64.3	(i) 69.0	(j) 81.1	(k) 67.4	(l) 78.7	(m) 65.8	(n) 75.9	(o) 6.7	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	95.5	73.7	63.3	90.5	72.4	61.8	85.8	70.9	33.7	81.6	32.4	78.4	31.1	76.1	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.3	(b) 17.2	(c) 14.7	(e) 8.5	(f) 91.6	(g) 9.7	(h) 2.5	(i) 1.5	(j) 93.4	(k) -4.1	(l) 94.9	(m) -9.5	(n) 96.3	(o) -16.6	(p) 98.2		
			DB													
			WB													
				8.5	91.6	9.7	2.5	1.5	93.4	-4.1	94.9	-9.5	96.3	-16.6	98.2	
				7.7	71.7	9.4	1.0	1.0	72.4	-4.6	73.0	-9.9	73.5	-16.7	74.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	33.8	35.1	42.9	48.0	57.5	62.0	66.5	66.9	59.4	51.4	41.1	37.1
	DBStd	13.72	9.40	9.71	7.59	6.42	6.75	6.84	5.44	6.30	5.97	6.67	7.16	7.47
	HDD50	2035	503	420	238	111	18	1	0	0	2	63	276	403
	HDD65	5674	967	837	684	510	240	139	48	56	188	423	718	864
	CDD50	2124	1	3	18	51	251	362	511	524	284	107	8	4
	CDD65	288	0	0	0	0	8	50	93	115	21	1	0	0
	CDH74	2129	0	0	1	3	69	348	688	873	133	14	0	0
	CDH80	506	0	0	0	0	2	64	163	259	15	3	0	0
(14) Wind	WSAvg	6.5	6.8	7.2	7.9	6.9	6.2	6.1	6.2	6.1	5.9	6.0	6.4	6.6
(15) Precipitation	PrecAvg	26.1	1.5	1.4	1.5	1.9	3.2	3.5	2.8	2.9	2.0	1.7	2.2	1.8
	PrecMax	38.4	3.7	4.1	5.4	4.7	8.3	7.1	6.5	6.6	5.4	4.5	4.1	4.5
	PrecMin	17.9	0.2	0.4	0.3	0.5	1.0	1.1	0.3	0.5	0.7	0.2	0.4	0.0
	PrecStd	4.3	0.7	0.7	0.9	1.1	1.4	1.5	1.5	1.4	0.9	1.1	1.0	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	61.8	70.3	72.3	79.1	85.6	88.3	91.6	82.2	74.8	60.6	57.4
		MCWB	48.2	50.3	54.9	55.4	62.0	68.3	68.1	68.3	63.1	60.4	52.3	50.8
	2%	DB	50.3	55.4	62.3	67.3	76.0	81.5	84.5	86.5	78.1	68.7	56.5	54.1
		MCWB	44.9	47.8	50.6	53.3	59.4	66.1	67.3	66.9	63.4	58.6	51.2	48.5
	5%	DB	47.5	51.2	57.3	63.2	72.8	78.2	81.1	82.3	74.0	64.0	53.5	50.7
		MCWB	43.1	44.9	47.9	51.2	57.9	64.5	65.8	65.9	62.0	56.6	49.3	46.3
	10%	DB	45.4	47.4	53.9	59.2	69.8	74.5	77.8	78.8	70.0	60.6	50.8	47.6
		MCWB	41.8	43.2	46.4	49.2	57.2	62.3	64.2	64.4	60.4	54.9	46.9	44.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	51.0	55.0	56.3	64.6	70.5	70.7	70.5	68.0	61.9	54.0	52.1
		MCDB	54.0	59.5	69.1	68.9	74.2	81.5	82.7	85.6	76.8	70.9	58.2	55.7
	2%	WB	46.0	48.4	51.2	54.3	61.5	68.0	69.0	68.7	64.8	59.6	52.1	49.1
		MCDB	49.8	55.2	60.0	65.4	71.2	78.8	80.2	82.8	73.2	67.0	55.5	53.2
	5%	WB	43.7	45.7	49.0	52.5	59.7	65.8	67.3	67.0	63.1	57.6	49.7	46.6
		MCDB	46.7	50.2	56.2	60.9	69.2	74.8	78.2	78.9	71.0	63.7	53.2	50.6
	10%	WB	42.0	43.6	47.1	50.5	58.0	63.7	65.5	65.5	61.3	55.5	47.0	44.2
		MCDB	45.2	47.6	53.3	57.9	67.5	72.7	75.4	76.2	69.1	59.5	50.4	47.3
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	11.4	12.7	14.1	15.9	14.5	15.7	16.8	15.1	13.4	10.0	8.0
		MCDBR	11.1	17.6	19.9	21.2	21.7	20.8	22.2	23.6	22.0	19.2	13.6	10.7
		MCWBR	7.8	11.1	10.6	10.2	9.8	8.6	9.2	8.4	9.3	10.2	9.5	7.8
	5% WB	MCDBR	9.8	15.4	17.5	18.2	18.2	19.0	19.1	20.1	18.9	17.9	12.2	10.5
		MCWBR	7.7	10.3	10.0	9.3	9.4	8.3	8.5	8.0	8.9	10.1	9.1	8.1
(40) Clear-Sky Solar Irradiance	taub		0.308	0.333	0.364	0.398	0.408	0.413	0.413	0.409	0.382	0.357	0.329	0.305
	taud		2.442	2.384	2.323	2.255	2.258	2.279	2.271	2.304	2.358	2.420	2.453	2.442
	Ebn at Noon		240	259	268	269	271	270	267	263	259	246	229	225
	Edh at Noon		20	27	34	40	41	41	41	37	32	26	20	18
(44) All-Sky Solar Radiation	RadAvg		294	538	913	1350	1595	1812	1702	1483	1095	661	341	234
	RadStd		41	88	133	187	172	145	156	142	122	69	41	38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	+0.61	+0.56	N/A	N/A	N/A	+36

Nomenclature: See separate page